

Cognitive Capacity and Plasticity

BDNF, COMT, KIBRA, nicotinic receptors, klotho, cognition and education scores, and neuronal plasticity throughout life

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

- 1. Summary 3
- 2. How to read this report 4
- 3. Polygenic scores 5
- 4. Key variants by topic 6
- 5. GWAS loci burden 8
- 6. Favourable 9
- 7. Unfavourable 10
- 8. Mechanisms of ageing 11
- 9. Practical suggestions 12
- 10. Limitations 13
- 11. Methods and sources 14
- 12. Annex: all variants analysed 15

1. Summary

Favourable. The strongest genetic factor of cognitive aging is absent: **APOE ε3/ε3** (without ε4) and TOMM40 rs2075650 A/A. Sérgio F is **MAPT H2/H2** (rs1052553 G/G), the tau haplotype associated with less risk of Parkinson's disease and tauopathies. It has **FADS1 rs174537 G/G** (good endogenous conversion of omega-3), **KIBRA T** (episodic memory) and DRD2 Taq1A without A1. The fluid intelligence scores are in the 78.7–percentile 79.3, at the upper limit of the typical, and that of schooling in the 65.7.

Unfavourable. BDNF Val66Met (Val/Met): less BDNF released with neuronal activity, so plasticity depends more on exercise, sleep and learning. There is 1 risk allele in CLU, PICALM and BIN1 (amyloid purification), and the Alzheimer's score with APOE is at percentile 90.8 at the expense of these non-APOE loci. The mild cognitive impairment score is at percentile 98.6, but it was developed in another population and has low reliability. It also has MTHFR C677T heterozygous (homocysteine and vessels) and **CHRNA5 D398N** heterozygous (only relevant with nicotine).

What does it mean? The genetic basis of cognition is good, and the genetic cognitive reserve is typical to high. The point to work is activity-dependent plasticity (BDNF Met) and protein clearance in the brain (non-APOE Alzheimer's loci). Both respond to the same levers: aerobic exercise, regular deep sleep, demanding learning and healthy vessels (LDL/ApoB low). COMT Val/Val gives a mixed profile: lower prefrontal dopamine at rest and better performance under pressure.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 17 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 14 of the chip, 3 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 1,905,817 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 8514 loci, 8097 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Cognition and education scores (in the sense of high percentile = more capacity) and cognitive aging (high percentile = more risk). Genetic education is an indirect indicator of cognitive reserve; it explains a small part of individual differences.

Cognition

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Fluid intelligence	PGS001919 (Privé 2022, 26 thousand variants)	25,428 26,145 (weighted coverage 97%)	79.3	TYPICAL	typical
Fluid intelligence (genomic)	PGS002135 (Privé 2022, Bayesian method, 903 thousand variants)	885,220 903,259 (weighted coverage 97%)	78.7	TYPICAL	typical

Schooling (cognitive reserve proxy)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Higher education	PGS002685 (Weissbrod 2022, Bayesian method, 980 thousand variants)	958,572 980,529 (weighted coverage 97%)	65.7	TYPICAL	typical
Higher education	PGS002587 (Weissbrod 2022, P+T, 514 variants)	459 514 (weighted coverage 88%)	29.4	TYPICAL	typical

- **Higher education:** Small score (only leading variants) that disagrees with the PGS002685 genomic; the genome, with 958 thousand variants, is the most reliable.

Cognitive aging

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Mild cognitive deficit	PGS018931 (Chen 2025, 37 thousand variants)	36,101 37,738 (weighted coverage 95%)	98.6	MORE LIKELY	unfavourable (low reliability)
Alzheimer's (including APOE)	PGS000811 (Najar 2021, 39 variants)	37 39 (weighted coverage 85%)	90.8	MORE LIKELY	unfavourable

- **Mild cognitive deficit:** Developed in a cohort of another population: uncertain transferability, low reliability.
- **Alzheimer's (including APOE):** Detailed in the Brain and Neurodegeneration report; here only as a context of cognitive aging.

4. Key variants by topic

Variants with effect described in the literature. The judgment refers to cognition and brain aging.

Plasticity and memory

BDNF is the main factor of synaptic plasticity and neurogenesis in the hippocampus. KIBRA (WWC1) participates in episodic memory and klotho protects synapses and myelin.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BDNF	rs6265	Val66Met	C/T	chip	T (1 copy/s)	unfavourable (light)	Val66Met: T allele (Met) reduces activity-dependent BDNF secretion; it is associated with slightly worse episodic memory and lower hippocampus, and the greater relative benefit of exercise	Moderate
WWC1 (KIBRA)	rs17070145		C/T	chip	T (1 copy/s)	favourable	T allele associated with better episodic memory (moderate effect, mixed replication)	Weak
KL (klotho)	rs9536314	KL-VS F352V	T/T	chip	G (0 copy/s)	neutral	KL-VS (F352V): 1 copy is associated with more klotho, better cognition and less decline; 2 copies without benefit	Moderate

Prefrontal dopamine, attention and acetylcholine

COMT degrades dopamine in the prefrontal cortex (working memory, attention). Nicotinic receptors (CHRNA) modulate attention and dopamine. DRD2/ANKK1 Taq1A affects the density of D2 receptors (error learning).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
COMT	rs4680	Val158Met	G/G	chip	A (0 copy/s)	mixed	Val158Met: Allele A (Met) reduces COMT activity and increases prefrontal dopamine (working memory ↑ at rest; more stress-sensitive). G/G (Val/Val): lower prefrontal dopamine, more stress resistance	Moderate
CHRNA5	rs16969968	D398N	G/A	chip	A (1 copy/s)	unfavourable (light)	D398N: lower sensitivity of the nicotinic receptor; dependence on nicotine ↑ (relevant only with nicotine)	Moderate
DRD2/ANKK1	rs1800497	Taq1A	G/G	chip	A (0 copy/s)	favourable	Taq1A: allele A (A1) is associated with less D2 receptors and less efficient error learning	Weak

Cognitive aging

Genes that weigh on cognitive aging: APOE (the strongest), TOMM40, MAPT (tau), CLU, PICALM and BIN1 (amyloid clearance and endocytosis) and MTHFR (homocysteine and cerebral vessels).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
APOE	rs429358		T/T	imputation DR2=0.99	C (0 copy/s)	favourable	Defines ε4 (C allele): the greatest genetic factor of cognitive decline and Alzheimer	Strong
APOE	rs7412		C/C	chip	T (0 copy/s)	neutral	Defines ε2 (T allele): protector for cognitive decline	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TOMM40	rs2075650		A/A	chip	G (0 copy/s)	favourable	TOMM40/APOE region (cognitive decline) [GWAS Catalog: Alzheimer's disease, G allele, OR 2.53, p=1E-295, PMID 20460622]	Moderate
MTHFR	rs1801133	C677T	G/A	chip	A (1 copy/s)	unfavourable (light)	C677T: higher homocysteine (vascular-cognitive risk factor)	Moderate
CLU	rs11136000		T/C	chip	C (1 copy/s)	unfavourable (light)	Clusterin (amyloid purification)	Moderate
PICALM	rs3851179		T/C	chip	T (1 copy/s)	unfavourable (light)	Endocytosis/depuration of amyloid [GWAS Catalog: Alzheimer's disease, T allele, OR 0.90, p=3E-48, PMID 35379992]	Moderate
BIN1	rs6733839		C/T	chip	T (1 copy/s)	unfavourable (light)	Tau and endocytosis [GWAS Catalog: Alzheimer's disease, proxy Alzheimer's disease or related of, allele T, OR 1.17, p=2E-139, PMID 42237039]	Moderate
MAPT (17q21)	rs1052553		G/G	chip	G (2 copy/s)	favourable	Haplotype H1/H2: the G-marked allele H2, associated with less risk of Parkinson's disease and tauopathies (progressive supranuclear paralysis); H1 is the risk haplotype	Moderate

Lipids and brain energy

FADS1 limits the endogenous production of EPA and DHA, the structural omega-3 of neuronal membranes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FADS1	rs174537		G/G	chip	T (0 copy/s)	favourable	The T allele reduces the activity of FADS1 (less EPA/DHA endogenous from ALA)	Moderate

Reference loci of cognition and schooling

Two replications of schooling and cognition, as an example. The sum of all loci is in section 5.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
POU3F2 (region)	rs9320913		C/A	imputation DR2=0.99	A (1 copy/s)	favourable	Replicated educational locus (cortical development) [GWAS Catalog: Educational attachment, allele A, beta 0.101 (unspecified unit), p=4E-10, PMID 23722424]	Moderate
FOXO3 (region)	rs2490272		C/C	imputation DR2=1.00	T (0 copy/s)	neutral	Cognition locus [GWAS Catalog: Intelligence, T allele, beta 7.441 z score increase, p=1E-13, PMID 28530673]	Weak

5. GWAS loci burden

Sum of all GWAS Catalog cognition loci ($p \leq 5 \times 10^{-8}$), each allele oriented towards more education, cognitive performance, memory or reaction time (slowest), and compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Schooling	6411 (6047)	6164 of 12094	6043	59.6	typical
Cognitive performance / intelligence	1987 (1941)	1888 of 3882	1935	38.7	typical
Memory	48 (47)	53 of 94	47	63.0	typical
Reaction time	68 (62)	70 of 124	62	78.8	typical

6. Favourable

- **APOE ε3/ε3** (rs429358 T/T imputed DR2 0.99; rs7412 C/C): without ε4. **Strong.**
- **TOMM40 rs2075650 A/A**: without the risk allele of the APOE/TOMM40 region. **Moderate.**
- **MAPT H2/H2** (rs1052553 G/G): the haplotype H2 is associated with less Parkinson and less progressive supranuclear palsy; H1 is the risk. **Moderate.**
- **FADS1 rs174537 G/G**: typical activity of desaturase. More EPA/DHA from precursors, with benefit to neuronal membranes. **Moderate.**
- **KIBRA rs17070145 C/T**: 1 copy of the T allele, linked to better episodic memory. **Weak.**
- **DRD2/ANKK1 Taq1A G/G**: without the A1 allele (less D2 receptors). **Weak.**
- **Fluid intelligence**: percentiles 79.3 and 78.7 (2 independent scores, ~97% coverage). Moderate for the direction, weak for the individual.
- **Schooling**: percentile 65.7 (genomic, Bayesian method); POU3F2rs9320913 with 1 allele of more education. **Weak moderate.**

7. Unfavourable

- **BDNF rs6265 C/T (Val/Met):** less activity-dependent BDNF secretion. On average, it is associated with slightly worse episodic memory and a smaller hippocampus. Met carriers earn more proportionally with exercise. **Moderate.**
- **CLU rs11136000 T/C, PICALM rs3851179 T/C and BIN1 rs6733839 C/T:** 1 risk allele in each (amyloid clearance, endocytosis and tau). Moderate (small effects, OR 1.1–1.2 by allele).
- **Alzheimer's score with APOE (39 variants):** percentile 90.8. As Sérgio F is $\epsilon 3/\epsilon 3$, the high percentile comes from the sum of non-APOE loci. The absolute risk remains dominated by the absence of $\epsilon 4$. **Moderate.**
- **Mild cognitive deficit:** percentile 98.6. It was developed in a cohort of another population, so the transferability is uncertain. **Weak.**
- **MTHFR C677T G/A:** homocysteine a little higher on average, a vascular-cognitive factor. Moderate for homocysteine, weak for cognition.
- **CHRNA5 D398N G/A:** more dependence and more effect of nicotine. It is only relevant if there is nicotine (tobacco, steam, tablets or "nootropic" adhesives). **Moderate.**
- **COMT Val158Met G/G (Val/Val):** mixed profile, is not a pure disadvantage. It has less prefrontal dopamine at rest (basework memory slightly worse) and better performance under stress. **Moderate.**
- **Neutral:** without KL-VS (klotho rs9536314 T/T, the common genotype).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Intercellular Communication	BDNF Val/Met (less BDNF activity); COMT Val/Val (prefrontal dopaminergic signal)	Unfavourable (mild) / mixed
Exhaustion of Stem Cells	BDNF Met: less support for hippocampal neurogenesis	Unfavourable (mild)
Protein Aggregation	APOE ε3/ε3 and MAPT H2/H2 (favourable); CLU, PICALM and BIN1 with 1 risk allele; Alzheimer's score at percentile 90.8	Mixed (favourable in APOE/MAPT core)
Lysosome dysfunction	PICALM and BIN1 (endocytosis and traffic to the lysosome): 1 allele risk each	Unfavourable (mild)
Glymphatic dysfunction	APOE ε3/ε3 (without ε4, which impairs glymphatic clearance); BDNF Met makes deep sleep more important	Favourable
Neuroinflammation	CLU (clusterin, complement) with 1 risk allele; without ε4 (less microglial activation)	Neutral
Vascular Aging	MTHFR C677T heterozygous (homocysteine and small cerebral vessels)	Unfavourable (mild)
Hematoencephalic Barrier dysfunction	Without ε4 (the ε4 accelerates the rupture of the barrier)	Favourable
Sleep disturbance	BDNF Met: memory consolidation depends more on sleep	Unfavourable (mild, indirect)
Redox imbalance	FADS1 G/G: neuronal membranes with more omega-3 long chain	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Aerobic exercise as the first plasticity tool (BDNF Met).** Zone 2 3-4×/week and 1-2 interval sessions, plus the planned strength training. It is the most robust stimulus of BDNF and volume of the hippocampus, and Met carriers gain proportionally more. Suggestion (not applied).
2. **Regular sleep of 7-9 h with deep sleep protected.** Memory consolidation and glymphatic clearance depend on it. The fixed time and the melatonin of the plan are maintained. Suggestion (not applied).
3. **Expensive and continuous learning for cognitive reserve:** a language, an instrument, a new motor competence or chess, with growing difficulty. It compensates for activity-dependent plasticity. Suggestion (not applied).
4. **Omega-3 3.2-4 g/day (already in the plan).** FADS1 G/G already converts well the precursors, and the direct EPA/DHA remains the most reliable way to reach the brain. There is no need to change. Suggestion (not applied).
5. **Homocysteine (MTHFR C677T).** Keep the 5-MTHF and B12 of the plan. Suggestion (not applied).
6. **Brain vessels:** LDL/ApoB as low as possible (the lower, the better), high fiber and stable glycemia. Non-APOE Alzheimer's loci and MTHFR add to vascular risk. Suggestion (not applied).
7. **Zero nicotine (CHRNA5 D398N):** without tobacco, steam, or "nootropic" nicotine pads. Suggestion (not applied).
8. **Protect the hearing.** Hearing loss is one of the greatest modifiable factors of cognitive decline. See the Vision and Hearing report. Suggestion (not applied).
9. **Cognitive baseline:** a standardized computerized cognitive test (memory, speed and executive functions) now and every 2-3 years, to follow the individual trajectory. Suggestion (not applied).
10. **Current medicine without genetic conflict with cognition:** Lithium 5 mg, rapamycin 4 mg of 6 in 6 weeks and telmisartan have a neuroprotective rationale (indirect evidence). Caffeine $\leq$ 200 mg only in the morning, as in the plan. Suggestion (not applied).

Not recommended: telomerase activators; "nootropics" with nicotine. Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed); the monthly senolytic is only fisetin + quercetin and only with normal ALT.

10. Limitations

- A genotyping chip with imputation does not read all variants. Rare variants of large effect (e.g. rare TREM2, PSEN1/2, APP) and repetitions (C9orf72) are missing.
- Cognition and schooling scores explain a small fraction of individual differences. Genetic education mixes cognitive capacity, personality and family environment. At the individual level they weigh more the environment, learning, sleep, exercise and vessels.
- The mild cognitive impairment score (PGS018931) was developed in another population, and its calibration for Sérgio F is uncertain.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- The effects of BDNF, COMT, KIBRA, DRD2 and CHRNA4/5 on cognition come from studies with mixed replication and small effects.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local statistical imputation (public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location (reference/alternative) are read from the Imputed VCF, which includes the SNPs of the chip. The published effect allele is compared with those alleles. If not coincided, the complement is used (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS001919 and PGS002135 (fluid intelligence; Privé 2022, PGS002685 and PGS002587 (schooling; Weissbrod 2022), PGS018931 (light cognitive deficit; Chen 2025 and PGS000811 (Alzheimer; Najjar 2021). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature (complete references available on request).

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Plasticity and memory	BDNF	rs6265	C/T	chip	T	1	strand +	Val/Met: less BDNF released with the activity
Plasticity and memory	WWC1 (KIBRA)	rs17070145	C/T	chip	T	1	strand +	1 copy of the protective allele
Plasticity and memory	KL (klotho)	rs9536314	T/T	chip	G	0	strand +	without KL-VS (common genome)
Prefrontal dopamine, attention and acetylcholine	COMT	rs4680	G/G	chip	A	0	strand +	Val/Val: lower prefrontal dopamine at rest (a slightly worse baseline working memory), better performance under stress
Prefrontal dopamine, attention and acetylcholine	CHRNA5	rs16969968	G/A	chip	A	1	strand +	1 copy of risk allele
Prefrontal dopamine, attention and acetylcholine	DRD2/ANKK1	rs1800497	G/G	chip	A	0	strand +	without the risk allele
Cognitive aging	APOE	rs429358	T/T	imputation DR2=0.99	C	0	strand +	without the risk allele
Cognitive aging	APOE	rs7412	C/C	chip	T	0	strand +	without the protective allele (common genotype)
Cognitive aging	TOMM40	rs2075650	A/A	chip	G	0	strand +	without the risk allele
Cognitive aging	MTHFR	rs1801133	G/A	chip	A	1	strand +	1 copy of risk allele
Cognitive aging	CLU	rs11136000	T/C	chip	C	1	strand +	1 copy of risk allele
Cognitive aging	PICALM	rs3851179	T/C	chip	T	1	strand +	1 copy of risk allele
Cognitive aging	BIN1	rs6733839	C/T	chip	T	1	strand +	1 copy of risk allele
Lipids and brain energy	FADS1	rs174537	G/G	chip	T	0	strand +	without the risk allele
Reference loci of cognition and schooling	POU3F2 (region)	rs9320913	C/A	imputation DR2=0.99	A	1	strand +	1 copy of the protective allele
Reference loci of cognition and schooling	FOXO3 (region)	rs2490272	C/C	imputation DR2=1.00	T	0	strand +	without the protective allele (common genotype)
Cognitive aging	MAPT (17q21)	rs1052553	G/G	chip	G	2	strand +	2 copies of the protective allele